

Mohamed Tarek Ismail

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Military Service: Completed

Education

M.Sc. in Computers and Systems Engineering (CSE) <i>Ain Shams University</i>	2025 – Now
B.Sc. in Electronics and Communication Engineering (ECE) <i>Ain Shams University</i>	2019 – 2024
Graduation Project: ASIC Physical Implementation of PULPino SoC Sponsored by ICpedia	
- Executed the PnR flow using Design Compiler, ICC2, StarRC, and PrimeTime for the PULPino RISC-V SoC. - Developed TCL scripts for Floor Planning , CTS , PnR , Parasitic extraction & Timing Sign off - Resolved MCMM timing violations via timing ECOs achieving 667 MHz timing closure.	
Undergraduate Research Assistant	supervised by prof. Hani Fikry ASU ICL
Design of low cost potentiostat for bacteria detection using ESP32.	

Experience

Verilog & FPGA Instructor – <i>National Telecommunications Institute</i>	Aug 2025 – Nov 2025
Delivered lectures and labs on RTL design and FPGA implementation of HW/SW systems.	
FPGA/ASIC Full-Time Intern – <i>Valeo</i>	Oct 2024 – Dec 2024
- Developed & Integrated SV behavioral reference model for SCALA 3 LiDAR verification. - Automated FPGA state extraction scripts to initialize reference models of DSP Filters aiding debugging.	
Design & Verification Trainee – <i>ICpedia</i>	Aug 2024 – Sep 2024
- SERDES Channel Equalization Techniques. - CDC & RDC Techniques - UVM & RAL	
FPGA/Firmware Intern – <i>SiliconArts</i>	Dec 2023 – Jun 2024
- Executed hardware/software integration of a Vortex RISC-V GPU on an FPGA platform. - Integrated the AXI XDMA PCIe subsystem for host-to-FPGA data transfer and incorporated XDMA software drivers into the Vortex testing code.	
Digital Design Summer Trainee – <i>Analog Devices</i>	Aug 2023 – Sep 2023
Designed the RTL for a matrix inversion sub-block for Least Squares Digital Predistortion targeting both FPGA and ASIC flows, achieving timing closure at 300 MHz.	

Projects

Tape-Out of a Quantized CNN ECG Classifier AMS chip on SkyWater 130 nm	American University in Cairo
- Implemented a tiny ECG arrhythmia classifier using HLS4ML. - Developed a pyUVM verification environment for end-to-end verification of the ECG inference pipeline. - Built a UVM-based HIL testing environment, reusing a PyUVM verification setup for FPGA targets. - Executed co-simulation of an ADC macro using Spicebind to verify analog-digital operation - Performed full ASIC PnR using LibreLane, meeting placement and routing closure under the 1 mm² floorplan.	
Design , Verification & FPGA implementation of a MIPI D-PHY layer	GitHub
- Design of a MIPI D-PHY PHY layer in SystemVerilog with clock and data lane CIL supporting HS, LP, and ULPS operating modes per the MIPI D-PHY specification. - Verified the design using a UVM testbench with a PPI UVC and SVA assertions verifying D-PHY timing specs. - Implemented the HS analog front-end on Xilinx FPGA using Xilinx IO primitives, synthesized and mapped via Vivado.	
SERDES Elastic Buffer Design & Verification	GitHub
- Designed the RTL for an Elastic Buffer using SystemVerilog running on 250 MHz. - Developed a UVM environment with Write and Read agents for asynchronous data domains, alongside an APB UVC that supports UVM RAL. - Developed a clock agent to model Spread Spectrum Clocking (SSC) - Verified data integrity and SKP Ordered Set handling across independent clock domains using Scoreboard & SVA - Developed functional coverage models to ensure compliance with USB 3 Gen1 specifications.	
Multichannel Router Verification (UVM & RAL)	
- Built a UVM Testbench incorporating multiple Rx/Tx packet, host bus for configuration clocking & reset UVCs. - Implemented constrained random stimulus generation and hardware register verification via RAL. - Developed a scoreboard to verify end-to-end packet routing and data integrity. - Developed Virtual sequences and system-level tests.	
AXI4 FIR Filter Custom IP Design & PetaLinux Integration on Zynq7000 PS	GitHub
- Designed a Verilog FIR IP (AXI4-Stream/AXI-Lite) integrated with the Zynq PS via AXI DMA. - Built a unified C++ test harness with a DMA shim for Verilator simulation and PetaLinux execution. - Authored BitBake recipes and device tree overlays for out-of-tree modules, UIO binding, and user-space deployment.	